

GENERAL SPECIFICATION G8.4

Louvres and Cladding

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G8.4 LOUVRES AND CLADDING

G8.4.1 Louvres

- (a) This Section specifies the requirements necessary to provide fixed louvres and frames, sealants; and bird and insect screens.
- (b) Louvres shall conform to Singapore Standard SS 212: A Aluminium Alloy Windows and the following standards:

Standard	Subject
BS EN 13030	Ventilation for Buildings - Terminals - Performance Testing of louvres subjected to simulated rain
BS EN 3506-1	Mechanical properties of corrosion-resistant stainless steel fasteners. Bolts, screws and studs
BS 6213 + A1	Selection of Construction Sealants
BS EN ISO 11600+A1	Silicone sealants
BS EN 1999-1-4 + A1 and BS EN 1999-1-1 + A2	Aluminium
BS EN 1991-1-4 + A1, BS EN 1991-1-1, BS EN 1991-1-7 + A1, BS EN 1991-1-3 + A1	Loading
BS 5250 + A1	Condensation in buildings

- (c) Refer to other relevant Standards and Codes of Practice to S.O.'s approval: Code of practice for the design of non-load bearing external vertical enclosures for buildings:

Standard	Subject
BS EN 573-3, BS EN 755-6, BS EN 755-3, BS EN 755-4, BS EN 755-2, BS EN 755-8, BS EN 755-7, BS EN 12020-2, BS EN 755-9, BS EN 486	Aluminium extrusions - wrought aluminium and aluminium alloys
BS EN 573-3	Aluminium and aluminium alloys - chemical composition and form of wrought products
BS EN 755 (BS EN 755-1, BS EN 755-2, BS EN 755-3, BS EN 755-4, BS EN 755-5, BS EN 755-6, BS EN 755-7, BS EN 755-8, BS EN 755-9)	Aluminium and aluminium alloys
BS 1161	Specification for aluminium alloys for structural purposes

Standard	Subject
PD 6484	Bimetallic corrosion
AAMA 2065	High-performance coating (PVDF) specification for exterior, performance requirements and test procedures for superior performing organic coatings on aluminium extrusions or panels
AS 1231	Aluminium and aluminium alloys. Anodic oxidation coatings
AS 3715	Metal finishing Thermoset powder coating for architectural applications of aluminium and aluminium alloys

- (d) Louvres shall be designed to permit passage of air velocity of 3m/s without blade vibration or noise with maximum static pressure loss of 12 Pa measured at 1.25m/s face velocity. The louvres shall have no rain water penetration at 1m/s air velocity intake.
- (e) Louvre units shall have a minimum 50 percent free area unless otherwise approved by the S.O..
- (f) Fixed louvre assembly shall consist of 150 mm (6 inch)-deep, 2.04 mm (0.081-inch)-thick, fixed extruded aluminium frame, invert V-shape blades spaced on 100-mm (4-inch) centers with 45-degree pitch, insect screen mounted on inside face, concealed mullions, interior duct collar, rated and tested in accordance with AMCA standards.
- (g) Pressure loss through the louvre assembly shall be no greater than 2.04 mm (0.1 inch) WC at 239 meters per minute velocity through the free area.
- (h) Bird and Insect Screen shall be interwoven wire mesh of galvanized steel, 1.5mm diameter wire, 10mm by 10mm open-weave, square design.
- (i) Fasteners and Anchors shall be stainless steel.
- (j) Flashings shall be of the same material as louvre frame.
- (k) Louvres shall be finished with siliconised polyester powder paint; colour to match adjacent wall finish. Finish shall be factory-applied, baked-on coating consisting of epoxy primer and silicone-modified polyester enamel topcoat with dry film thickness of not less than 0.005 mm for primer and 0.04 mm for topcoat.
- (l) Louvre assembly shall be installed in accordance with manufacturer's instructions. Louvres shall be installed level and plumb. Secure louvres in framed opening with concealed fasteners, removable for maintenance purposes. Bird and insect screening shall be installed to the interior of louvres.
- (m) Where indicated, prepare louvres to receive ductwork attachment required in accordance with specifications for metal HVAC ductwork.

G8.4.2 Aluminium Curtain Wall

- (a) This Section specifies the requirements necessary to furnish and install aluminum tube frame aluminum panel curtain wall system including:
 - (i) Aluminum framing components;
 - (ii) Casement windows for FAP;
 - (iii) Integral louvres; and
 - (iv) Structural design of curtain wall assembly including framing members, glass, reinforcements, and connections to building structural framing to resist working forces.
- (b) Aluminum panel curtain wall systems shall conform to the following specifications:

Standard	Subject
AAMA 501.2	Field Check of Metal Storefronts, Curtain Walls, and Sloped Glazing Systems for Water Leakage
AAMA 502	Voluntary Specification for Field Testing of Windows and Sliding Glass Doors
AAMA 503	Voluntary Specification for Field Testing of Metal Storefronts, Curtain Wall and Sloped Glazing Systems
ASCE 7	Minimum Design Loads for Buildings and Other Structures
ASTM E1105	Standard Test Method for Field Determination of Water Penetration of Installed Exterior Windows, Curtain Walls, and Doors by Uniform or Cyclic Static Air Pressure Difference
BCA Approved Document – Acceptable Solutions. Latest version	

- (c) Design of Wall Systems shall comply with Euro codes and the following criteria:
 - (i) Wind Loads: components shall be designed and sized to withstand dead loads and live loads caused by positive and negative wind loads acting normal to plane of wall, including building corners.
 - (ii) Seismic Loads: components shall be designed and sized to withstand seismic loads and sway displacement as calculated in accordance with applicable code. Curtain wall system shall comply with safety barrier load requirements in BCA Approved Document 2013 – Acceptable Solutions.
 - (iii) System Assembly shall accommodate the following without damage to system, components, or deterioration of seals:
 - 1) Movement within system;
 - 2) Movement between system and perimeter framing components;
 - 3) Dynamic loading and release of loads;

- 4) Deflection of structural support framing;
 - 5) Tolerance of supporting components;
 - 6) Shortening of building concrete structural columns;
 - 7) Creep of concrete structural members;
 - 8) Interstory drift; and
 - 9) Mid-span slab edge deflection.
- (iv) No water leakage shall be permitted, when measured in accordance with ASTM E331 with test pressure difference of 20 percent of design pressure, with minimum differential of 2.86 pound-foot per square foot and maximum of 12 pound-foot per square foot.
- (v) Curtain wall assembly shall be designed and fabricated to allow for expansion and contraction within the assembly components caused by cycling temperature range of 170 degrees F over 12-hour period without causing detrimental affect to system components.
- (vi) System Internal Drainage shall drain water entering joints, condensation occurring in glazing channels, or migrating moisture occurring within system, to exterior by weep drainage network.
- (vii) Vibration harmonics, wind whistles, noises caused by thermal movement, thermal movement transmitted to other building elements, loosening, weakening, or fracturing of attachments or components of system shall not be permitted.
- (viii) Deflection requirements shall be in accordance with the applicable building code, or as a minimum, $L/180$ for wind load (but not less than 49 kg/sq m).
- (d) Aluminium panel curtain wall system as designed by the manufacturer shall be a complete system. All components of the system shall be supplied by the same manufacturer. Aluminum panel curtain wall, matching stops and glass retaining member of sufficient size and strength to provide bite on Fire Access Panels, drainage holes, deflector plates and internal flashings to accommodate internal weep drainage system; internal mullion baffles to eliminate stack effect air movement within internal spaces.
- (e) Curtain wall shall consist of tubular aluminum sections with self-supporting framing, factory prefinished, glass and glazing, insulated metal panel spandrel infill, and louvres; related joint sealers, flashings, anchorage and attachment devices. Wall system shall be reglazable from interior.
- (f) Fasteners shall be stainless steel.
- (g) FAP casement windows shall be openable from external side
- (h) Aluminium Surfaces shall be powder coated to match existing building curtain wall mullion colour.
- (i) Bituminous paint shall be applied to concealed aluminium and steel surfaces in contact with cementitious or dissimilar materials.
- (j) Finish shall be applied so no natural aluminium is visible in completed assemblies, including joint edges.

- (k) Tolerances shall not exceed a maximum variation from plumb: 0.06 inches every 3 feet noncumulative or 0.5 inches per 100 feet, whichever is less. Maximum Misalignment of Two Adjoining Members Abutting in Plane: 1/32 inch. Sealant Space Between Curtain Wall Mullions and Adjacent Construction: Maximum of 3/4 inch and minimum of 1/4 inch.

G8.4.3 Quality Assurance

- (a) Manufacturer for products specified in this Section shall be a company specialising in the products with minimum 3-years documented experience and with service facilities within 100 miles of project.
- (b) Installation shall be undertaken by a company specialising in performing similar work with minimum 3-years documented experience and approved by the manufacturer.
- (c) Design structural support framing components under direct supervision of a professional engineer experienced in design of this work and licensed in the jurisdiction where project is located.
- (d) Preconstruction Conference: convene conference a minimum 2 weeks prior to commencing work of this Section.

G8.4.4 Submittals

- (a) The following submittals shall be provided to the S.O. within 4 weeks of Contract award:
 - (i) Product data on preassembled louvres describing design characteristics, maximum recommended air velocity, free area, materials, and finishes.
 - (ii) Product Data of curtain wall systems providing component dimensions, describe components within assembly, anchorage and fasteners, glass and infill, internal drainage details, and water flow drainage diagrams.
 - (iii) Shop Drawings: Indicate louvre layout, elevations, dimensions, and tolerances; large-scale details of louvre configuration; screening; and frames.
 - (iv) Manufacturers test data indicating pressure loss through all louvres.
 - (v) Samples, Louvre Material: 150 mm by 150 mm in size, illustrating finish and colour of exterior and interior pre-finish.
 - (vi) Shop Drawings:
 - 1) Indicate system dimensions layout, framed opening requirements and tolerances, anticipated deflection under load, affected related work, weep drainage network, expansion and contraction joint location and details, and field welding required; and
 - 2) Include shop and field sealants by manufacture and product name, and locate on drawings. Show sealant joint sizes, including tolerances and maximum/minimum joint sizes required.
 - (vii) Samples: Submit two 150 mm by 150 mm in size indicating prefinished finish and colour of exterior and interior pre-finish, specified glass, insulated infill panels, glazing materials illustrating edge and corner.

- (viii) Structural Calculations: Indicate framing member structural and physical characteristics, calculations, dimensional limitations stamped by professional engineer registered in jurisdiction where project is located.
- (ix) Test Reports: Indicate substantiating engineering data, test results of previous tests by independent laboratory which purport to meet performance criteria, and other supportive data.
- (x) Manufacturer's Certificate: Certify products meet or exceed specified requirements.
- (xi) Installation Data: manufacturer's written installation instructions and details.
- (xii) Maintenance Manual: Provide an assembled and bound maintenance manual, describing the materials, devices, and procedures to follow in cleaning and maintaining the glazed aluminum curtain wall components. Include manufacturer's brochures describing the actual materials used in the work, including metal alloys, finishes, glass, sealants, gaskets, and other major components.

G8.4.5 Mock-Ups

- (a) Construct mockup, 1m by 4m mockup including intermediate mullion, corner mullion, sill muntins, vision glass light, and insulated infill glass types. Assemble to illustrate component assembly including glazing materials, weep drainage system, attachments, anchors, and perimeter sealant.

G8.4.6 Field Quality Control

- (a) Field Testing: Perform a field check of the mock-up for water penetration, in accordance with the procedure described in AAMA 501.2. Test wall area after glazing, but before installation of insulation or interior finish. Correct and retest any discovered leakage. If excessive leakage is discovered, up to 3,000 square feet of additional wall area shall be tested in areas selected by S.O., at no additional cost.
- (b) Perform one test each at 10, 50, and 80 percent of curtain wall completion, with repeat tests when failures occur. When testing results in leakage, eliminate causes of leaks and retest until no leaks occur.

G8.4.7 Warranties

- (a) Contractor shall provide a Warranty from the manufacturer of 10-year extended warranty in writing, with the Board named as beneficiary. Warranty shall provide for correction, or at the option of the Board, removal and replacement of louvre or glazed aluminium curtain wall found defective in relation to design criteria during the stated warranty period after Date of Substantial Completion.
- (b) Contractor shall repair or replace window wall components which have failed to provide watertight and airtight joints for any reason or which appear to have failed in adhesion, cohesion, abrasion resistance, migration resistance, stain resistance, general durability, or other form of apparent deterioration (excluding inherent qualities and limitations clearly specified in the manufacturer's data which was submitted).

- (c) Complete installation shall be warranted jointly and severally, on a single document by the materials manufacturer and applicator, against defects of materials and workmanship as defined on the warranty.
- (d) Coatings Warranty shall be provided in an approved form of guarantee that the paint coating or anodized aluminium will not develop excessive fading or excessive non-uniformity of colour or shade and will not crack, peel, pit, or corrode, within limits defined as follows:
 - (i) Excessive Fading: means a change in appearance which is perceptible and objectionable as determined by S.O. when viewed visually in comparison with the original acceptable colour range.
 - (ii) Excessive Nonuniformity: means non-uniform fading during the period of the warranty to the extent that adjacent panels have a colour difference greater than the original acceptable colour range.
 - (iii) Will Not Crack, Peel, Pit, or Otherwise Corrode: means there shall be no cracking, peeling, pitting, or other type of corrosion discernible from a distance of 10 feet resulting from the natural elements in atmosphere.
- (e) Before starting work, provide sample copy of warranties to be provided. Upon completion and acceptance of work required by this Section, submit an executed copy of warranties.

END OF SECTION